

Bear Model – Technical Documentation

Generated from `bear/bear_features.csv` and `bear/targets.csv`. Reproduce with `python -m bear.make_docs`.

1. Overview

- **Objective:** estimate a binary **12-month rolling correction**, $y_t = \mathbf{1}\{\text{MDD}_t \leq -10\%\}$, i.e. any drawdown deeper than 10% over the rolling 12-month forward window.
- **Model:** weight-constrained binary logistic on standardized factors, signs fixed to economic priors, calibrated to the base rate.
- **Horizon:** 12 months (rolling forward window).
- **Sample:** 1910-01-31 → 2026-06-30 (1397 complete monthly observations).
- **Historical base rate:** 20.8%.
- **Current reading (2026-06-30):** $P(\text{event}) = 16.6\%$.
- **Inference:** Newey-West HAC, max lag = 12 months.

2. Feature Engineering

All raw series are monthly (daily/weekly series are sampled to month-end), shifted forward by their real-world publication lag to prevent look-ahead, and transformed to stationary form. Each factor is then standardized on the training sample, $\tilde{x}_{i,t} = (x_{i,t} - \mu_i) / \sigma_i$.

#	Factor	Raw series	Transformation	Rationale
1	<code>infl_zscore_120m</code>	—	—	—
2	<code>infl_yoy</code>	—	—	—
3	<code>spx_vs_10ma</code>	S&P 500 price (monthly close)	% deviation from 10-month MA ($\approx$ 200-day)	Trend deterioration; price below the long MA is risk-off (Moskowitz-Ooi-Pedersen 2012).
4	<code>cape_z_120m</code>	—	—	—

Forward drawdown target (rolling window):

$$\text{MDD}_t = \min_{t < u \leq t+12} \left(\frac{P_u}{\max_{t < v \leq u} P_v} - 1 \right)$$

3. Constrained Logistic Model

Link / specification:

$$\hat{p}_t = \Pr(y_t = 1) = \sigma(z_t) = \frac{1}{1 + e^{-z_t}}$$

$$z_t = \beta_0 + \sum_{i=1}^4 \beta_i \tilde{x}_{i,t}$$

Estimation: coefficients are written as $\beta_i = \text{sign}_i \cdot \gamma_i$ with $\gamma_i \geq 0$ (signs fixed to economic priors), fitted by maximizing the Bernoulli likelihood subject to per-factor weight bounds:

$$w_i = \frac{|\beta_i|}{\sum_j |\beta_j|}, \quad 0\% \leq w_i \leq 100\%$$

Fitted model:

$$z_t = -1.625 + 1.139 \tilde{x}_1 - 1.195 \tilde{x}_2 - 0.480 \tilde{x}_3 + 0.378 \tilde{x}_4$$

Fitted coefficients (sorted by weight):

Factor	Coefficient β_i	Weight w_i	p (HAC)
infl_yoy	-1.1947	37.4%	0.002 *
infl_zscore_120m	+1.1392	35.7%	0.000 *
spx_vs_10ma	-0.4801	15.0%	0.016 *
cape_z_120m	+0.3775	11.8%	0.114
intercept	-1.6255	—	—

* p<0.05 · . p<0.10 (Newey-West HAC, max lag = 12).

4. Regression Performance

4.1 Specification

- **Form:** binary logistic, identity-of-weights constraint $w_i \in [0\%, 100\%]$, 4 factors, signs fixed.
- **Autocorrelation:** the 12-month rolling target overlaps across consecutive months (consecutive observations share 11/12 of their window). Standard errors are Newey-West HAC-corrected with a Bartlett kernel and max lag = 12 months.
- **Calibration:** fitted by natural-weight likelihood, so the output is calibrated to the 20.8% base rate.

4.2 Sensitivity

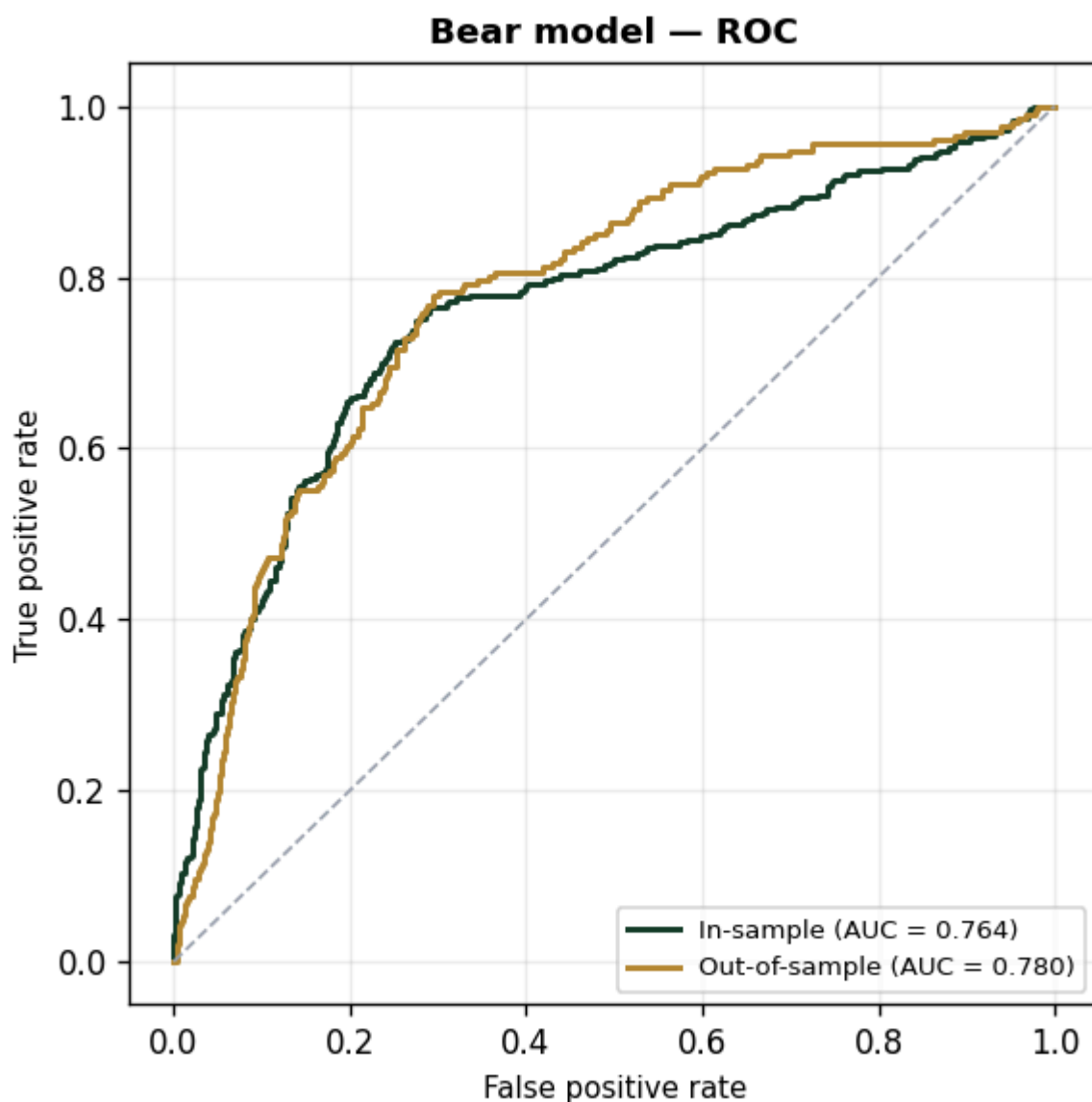
Marginal effect of a **+1 standard-deviation** move in each factor on the model output, evaluated at the current reading ($\partial \hat{y} = \hat{y}(1 - \hat{y}) \beta_i$, $\hat{y} = 0.166$):

Factor	β_i	Weight	Δ output / +1 SD	p (HAC)	Push
CPI inflation, YoY %	-1.1947	37.4%	-0.1650 (-16.50 pp)	0.002	Bullish
CPI inflation, 10yr z-score	+1.1392	35.7%	+0.1574 (+15.74 pp)	0.000	Bearish
S&P 500 vs 10-month MA	-0.4801	15.0%	-0.0663 (-6.63 pp)	0.016	Bullish
Shiller CAPE, 10yr z-score	+0.3775	11.8%	+0.0522 (+5.22 pp)	0.114	Bearish

4.3 Area Under the Curve (AUC)

Discrimination is measured against the realized event (*Bear event: 12-month drawdown > 20%*); the predicted probability is the ranking score.

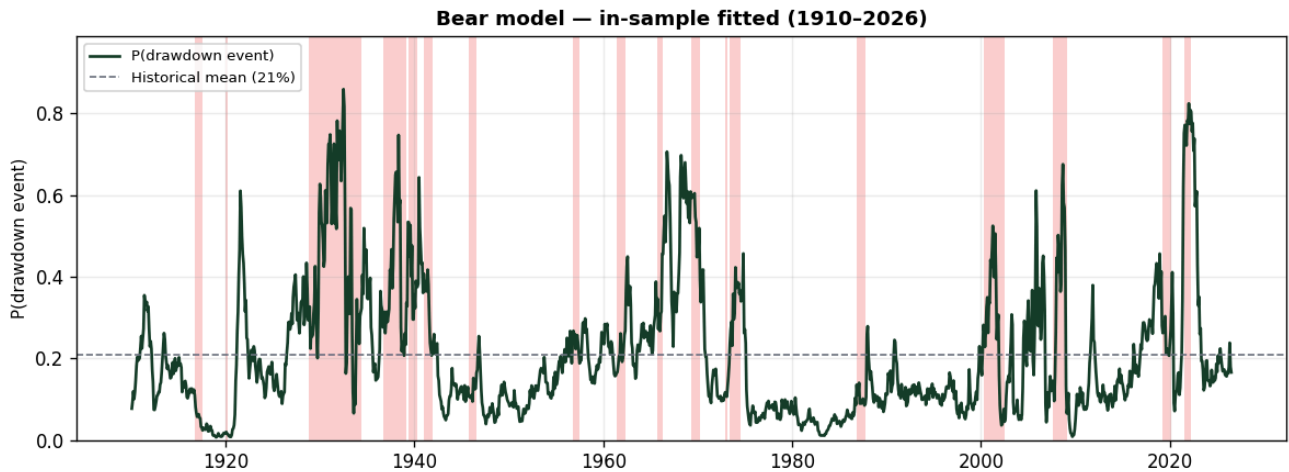
Sample	AUC	N	Events
In-sample	0.764	1397	290
Out-of-sample (walk-forward)	0.780	1097	207



5. Charts

5.1 In-sample fit

Fitted values across the full sample (parameters estimated on the full sample). Shaded spans mark the realized rolling-window event.



5.2 Out-of-sample (walk-forward)

Expanding-window estimate: at each month the model is re-fit on prior data only, then predicts that month (no look-ahead).



6. Appendix — Realized Bear Events

The 290 event-months in the AUC sample (signal months whose 12-month forward drawdown exceeded 20%) group into **19 distinct episodes**. *Start/End* are the first and last signal months of each run; *Worst drawdown* is the deepest 12-month forward drawdown observed during the episode.

#	Start	End	Signal months	Worst drawdown
1	1916-09	1917-07	11	-28.9%
2	1919-12	1920-03	4	-22.9%
3	1928-10	1934-05	68	-71.2%
4	1936-09	1939-02	30	-52.0%
5	1939-05	1940-04	12	-31.9%
6	1940-12	1941-11	12	-28.7%
7	1945-09	1946-07	11	-27.7%
8	1956-10	1957-06	9	-20.7%
9	1961-05	1962-04	12	-28.0%
10	1965-08	1966-03	8	-22.2%
11	1969-04	1970-03	12	-32.7%
12	1972-11	1973-02	4	-23.4%
13	1973-05	1974-07	15	-43.0%
14	1981-03	1981-03	1	-21.4%
15	1986-10	1987-09	12	-33.5%
16	2000-03	2002-06	28	-34.7%
17	2007-07	2009-01	19	-52.6%
18	2019-03	2020-02	12	-33.9%
19	2021-06	2022-03	10	-25.4%
		Total	290	